

The Concrete Advancement Network (CAN)

A consortium of academic and industrial researchers
engaged in bringing cutting-edge technologies to the
cement and concrete materials for the built environment

Mission and Vision

- **Mission:** To develop innovative concrete materials and processing technologies that will boost performance, reduce embodied carbon, reduce waste, and translate promising outcomes of university projects into tangible impacts for member companies while developing a diverse and skilled science and engineering workforce for the industry at large.
- **Vision:** Implementation of the technologies developed through CAN will enable the next level of performance in construction while conserving natural resources, reducing the industry's carbon footprint, and providing a value-added proposition for concrete in a circular economy.

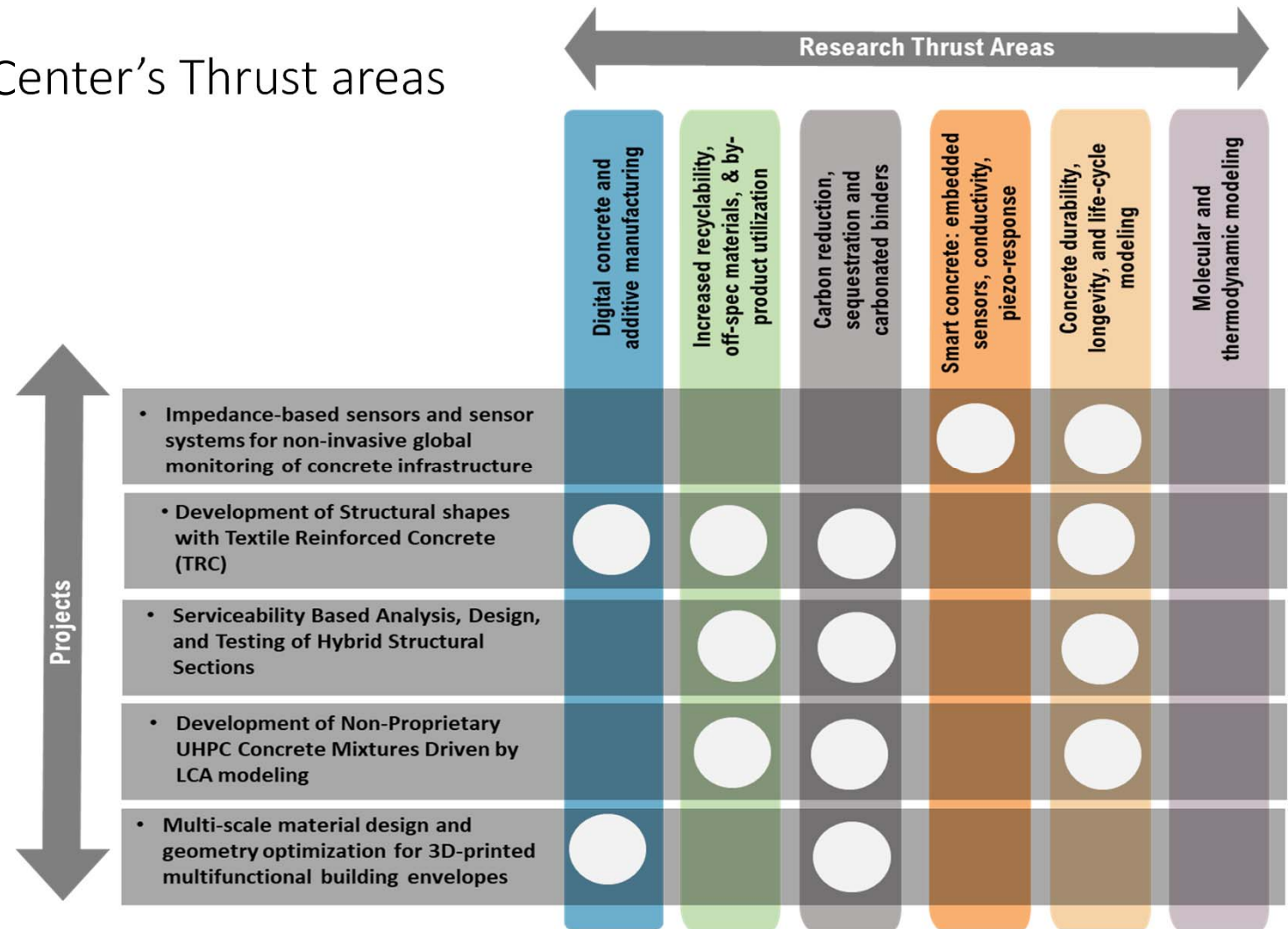
Four Convergent and Interrelated themes

- Contributions to novel and sustainable materials development,
- Use-and-performance inspired design;
- Advances in service-life enhancement, and
- Materials informatics, including machine learning and artificial intelligence.

Stakeholders and Industrial Partners

Industry Segments	Description
Owners, Contractors, testing labs, design, and consulting	Specifiers, end-users, expert, and testing companies
Trade and non-profit organizations, state and federal agencies	Standards and specifications defining bodies
Manufacturers of bulk ingredients: aggregates, SCM and by-products including off-spec minerals; Green alternatives	Bulk of raw materials for concrete production, driving sustainability initiatives and green solutions
Cement, concrete, and precast concrete manufacturers	Main industry segment and potential core members
Value added specialty Products Chemical admixture, fibers, and nanomaterials	Leading segment for innovation and sustainability

ASU Projects vs. Center's Thrust areas



What constitutes the areas of core competence for the ASU team?

- There are at least three major previous projects funded in each of these disciplines we promote
- There are at least 12 peer reviewed papers published in each of the subject areas by the ASU team
- There are at least three Ph.D. students who have graduated in each of the disciplines presented in the following areas
- The laboratory facilities are quite well developed and ready to initiate the projects

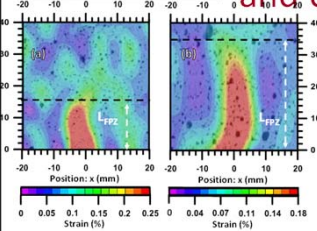
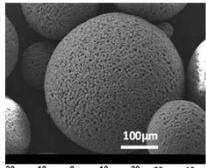
Materials Science of Infrastructural Materials

Multiple funded projects in all these areas – NSF, DOE, DOT, Private industry, over the past 5 years; 60 publications

Developing Ultra High-Performance Concretes

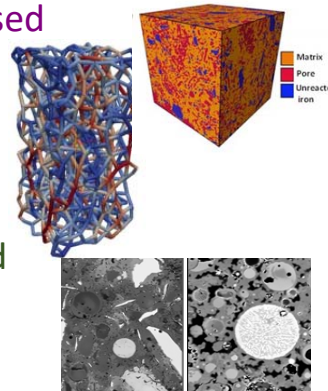
Limestone in concretes
Phase change materials
for thermal efficiency
and crack control

Ultra high
performance,
ductile bridge
concretes



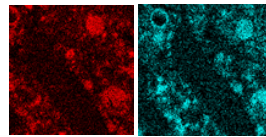
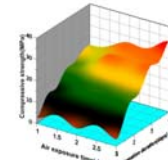
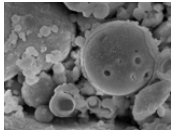
Microstructure-Guided Material Design

First-principles based
material design to
predict properties
and design new
materials
Concrete, soils and
biomaterials



Carbon-neutral/negative materials

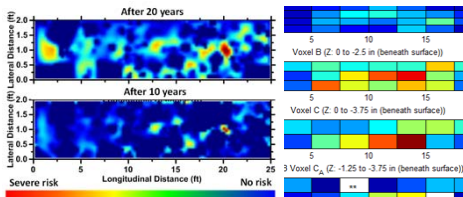
Binders from 100% waste
materials (e.g., fly ash)
Carbonation of Metallic
powders
Upscaling
carbonation
for beneficial
products



CO₂ Sequestration by
mineralization

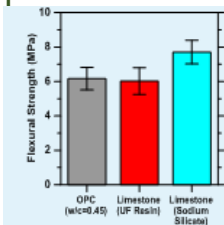
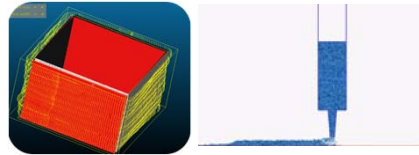
Sensors/Sensor Systems

Risk management/decision making
for corrosion-prone structures
Sensing systems for sub-surface
detection



3D Printing Structural Elements

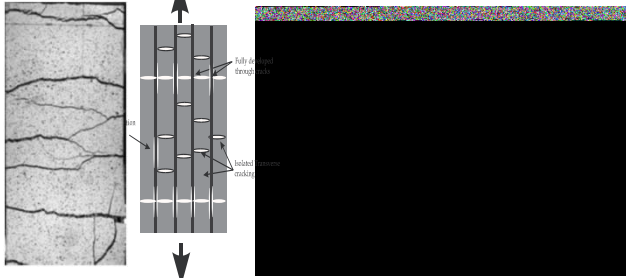
Linking
material and
processes
for efficient
printing of
controlled
architectures



Structural Aspects of Infrastructural Materials

Multiple funded projects in all these areas –DOT, SRP, BASF, AECOM – 60+ publications over the past 5 years

Ultra High-Perf. Concretes



Constitutive relations for structural analysis, and crack width control, High ductility and tension stiffening effects
Ultra high performance, ductile elements for bridge construction

Serviceability-based Design guidelines

First-principles based material design to predict properties and control design based on stiffness, ductility, allowable deflection, crack width new materials
Durability modeling



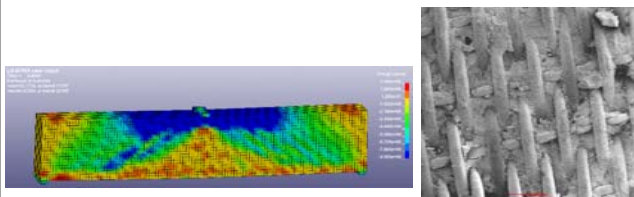
Ultra ductility and Thermally efficient structural systems

Textile Reinforced Concrete for new and repair systems



Supplementary Reinforcing systems

Modeling of the synergy effects of hybrid reinforcement for shear and serviceability



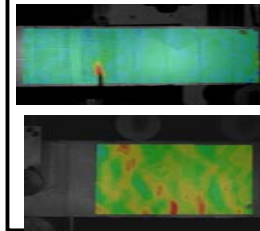
Structural Elements to Tensile Apps

Linking material and processes for efficient members of construction domain to compete with wood and steel



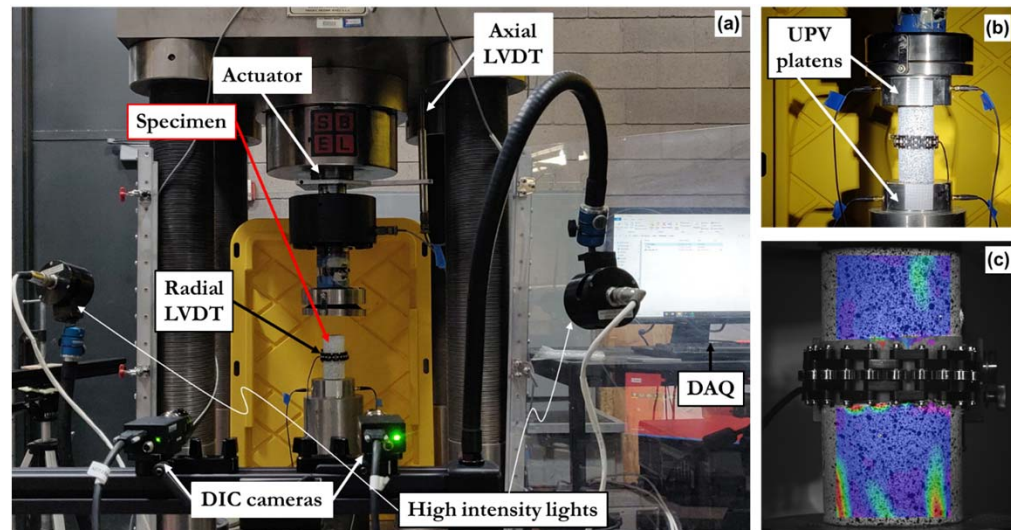
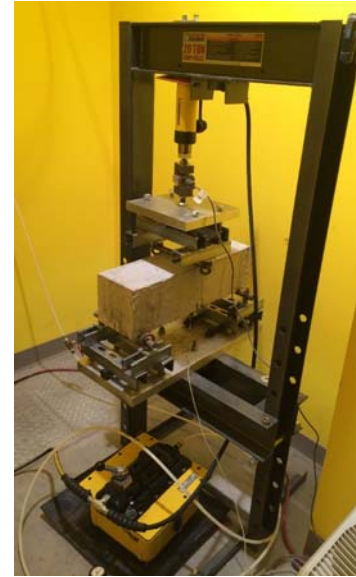
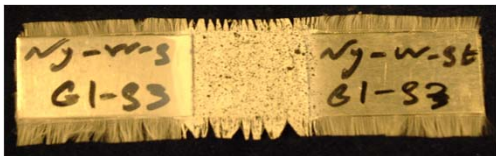
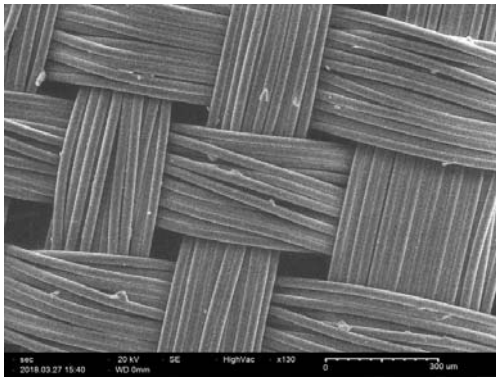
QC and Specs for Structural Concrete

Statistical process Control, Risk management /decision making for QC specifications.
Analysis of outliers



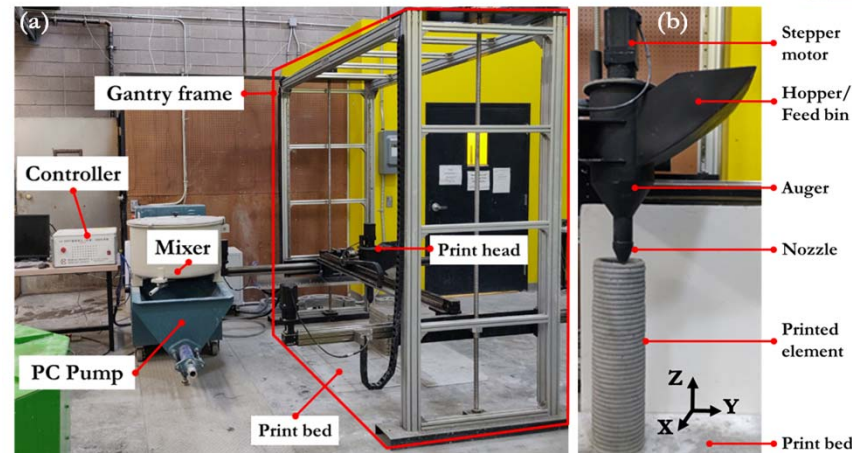
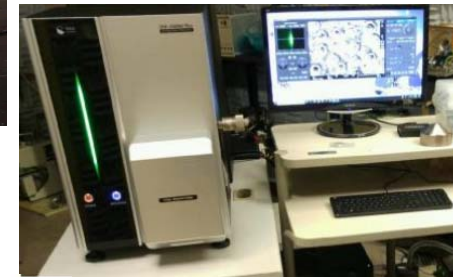
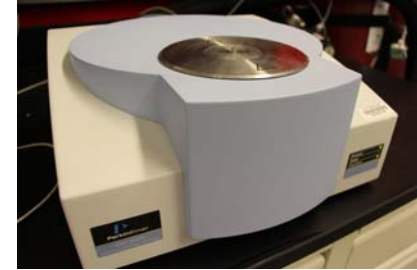
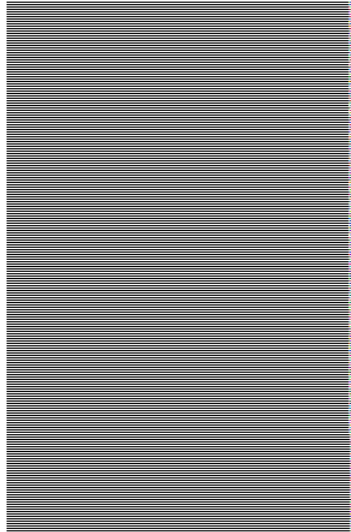
Projects and Capabilities at ASU

- Mechanical Testing, Constitutive modeling,
- Servohydraulic testing systems, impact, high speed testing, creep, high temperature



Capabilities

- MIP, SEM-EDX, XRD, AFM, Nano indentation, Micro-indentation, Focused Ion Beam, BET, Pycnometry, 3-D printing, microscopy, DTA/TGA

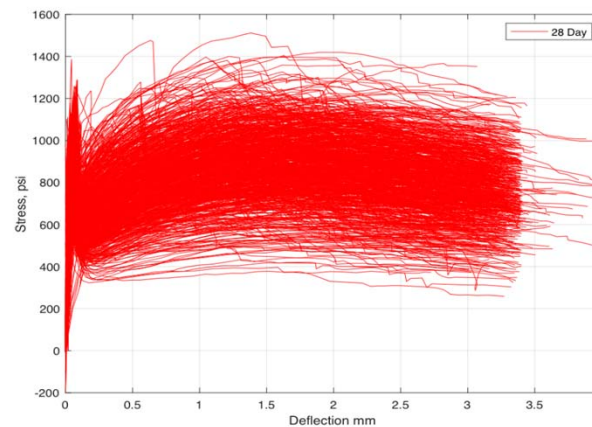


ASU Structural-Mechanics Research Lab

Develop Sustainable Materials, analyze using solid mechanics, and materials constitutive formulations, Structural components, Full scale testing and Modeling

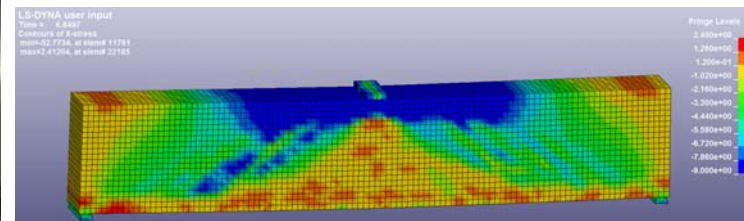
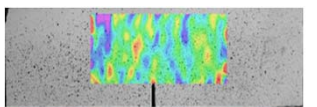
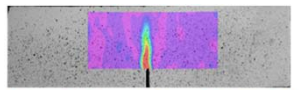
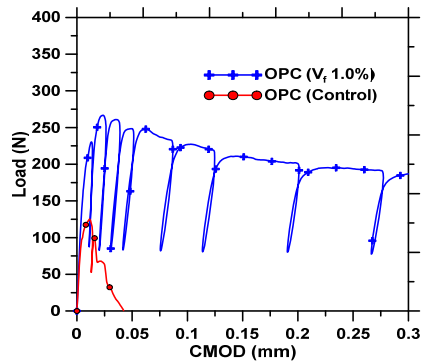
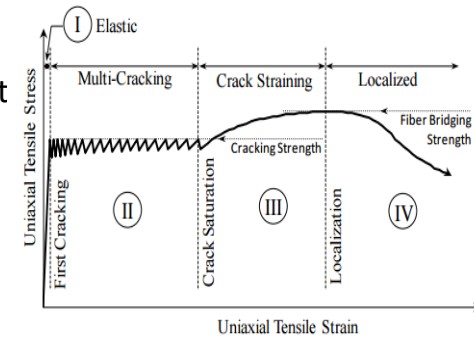
Promote innovative and sustainable Materials.

Innovative solutions for using composite materials for transportation, water treatment facilities, elevated slabs, tunnel lining, Thin section composites, Structural Shapes



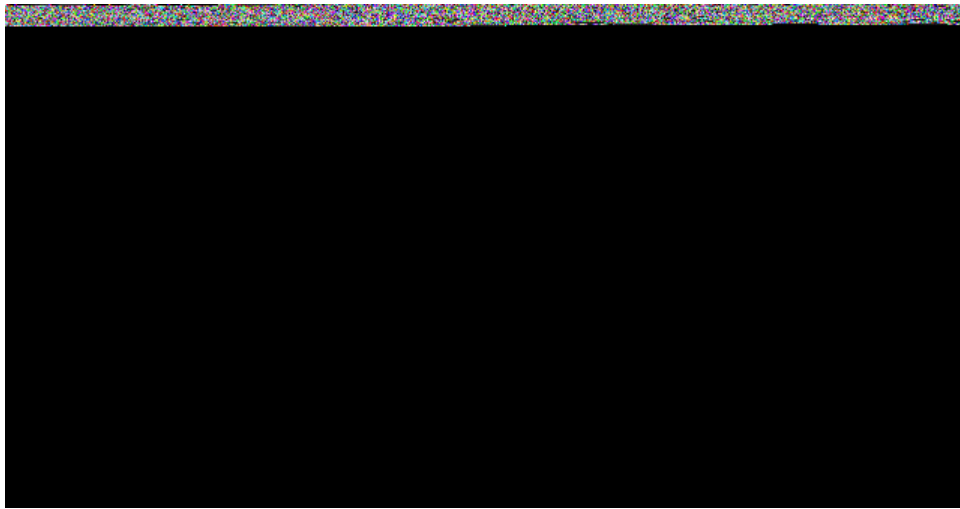
Ultra high-Performance Concrete

- Designing non-proprietary formulations for UHPC
 - Utilize Local materials, concrete with 24000 psi strength and 1000 times more ductility
 - Particle packing methods and Rheological properties
 - Experiments, Design guides, and simulations

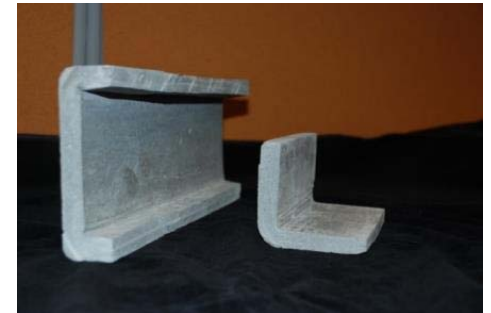
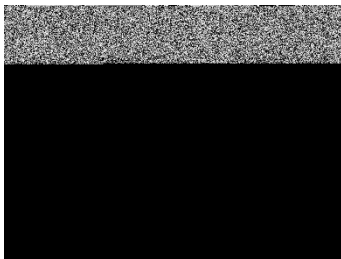


Structural Shapes using Textile Cement Composites

- Using an automated pultrusion system, full scale structural shapes composed of TRC laminates are efficiently manufactured.
- Competitive to wood and Light gage steel sections

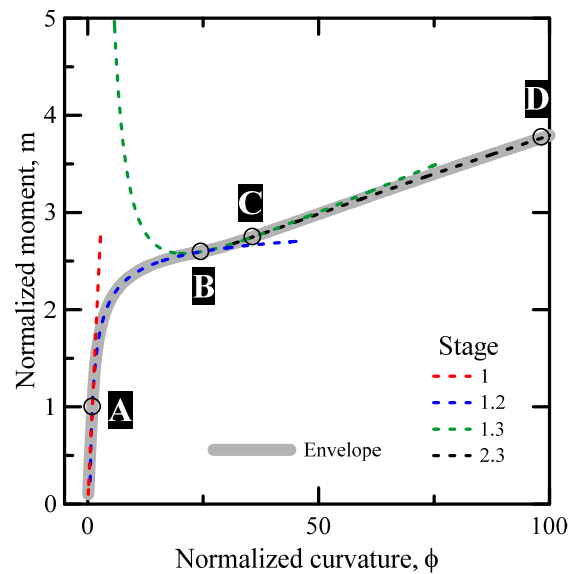


c Diagram



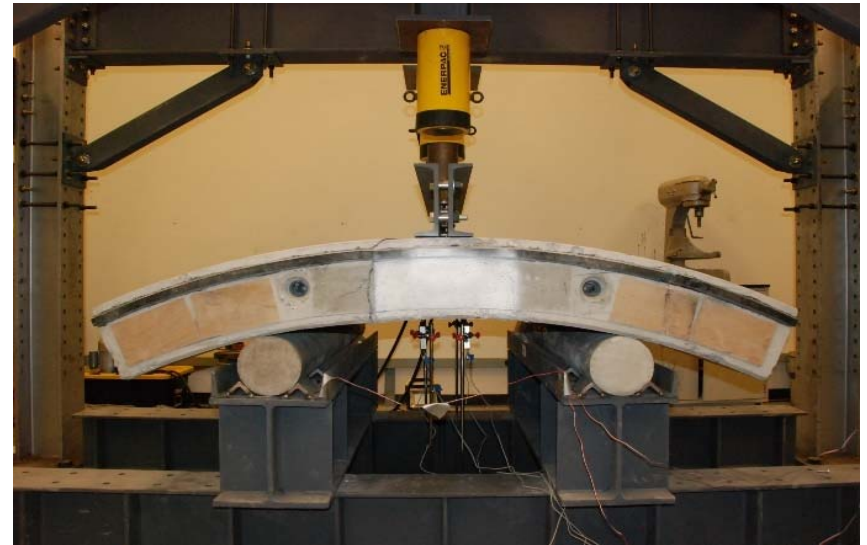
Development of Structural shapes with Textile Reinforced Concrete (TRC)

- **Team:** Barzin Mobasher, Steve Schaef (Master Builders Solutions)



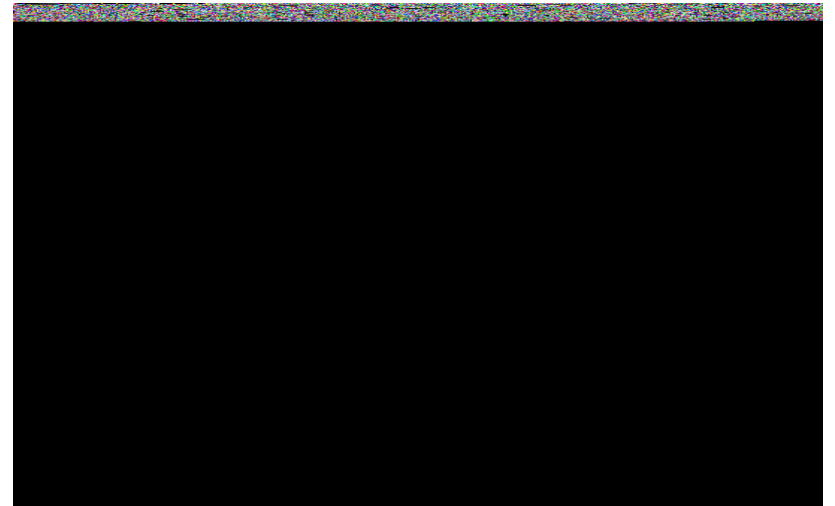
Serviceability Based Analysis, Design, and Testing of Hybrid Structural Sections

- **Team:** Barzin Mobasher, Mehdi Bakhshi (AECOM), William Geers (Bekaert), Amir Bonakdar (Euclid Chemicals), Steve Schaef (Master Builders Solutions)



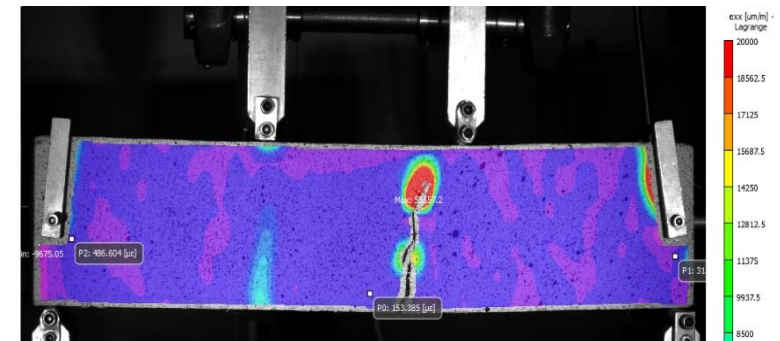
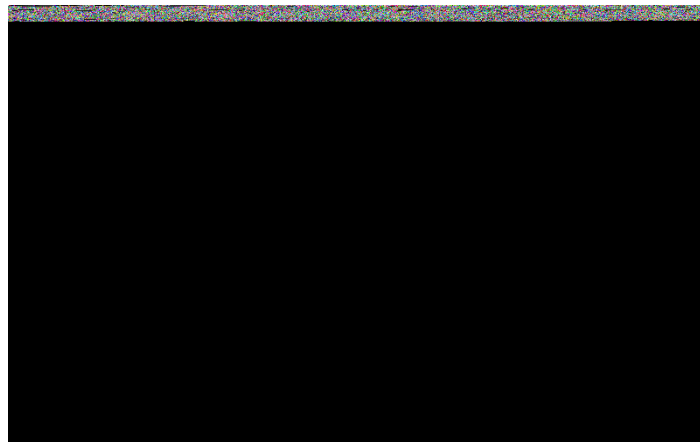
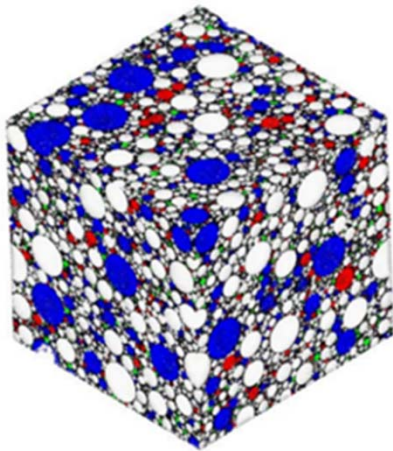
Impedance-based sensors and sensor systems for non-invasive global monitoring of concrete infrastructure

- **Team:** Narayanan Neithalath, Don Colosimo and John Hewitt (Transtech Systems)
- Modeling the electromagnetic field in concrete towards developing 1D/2D surface electrode arrays
- electrode models and tomographic algorithm development and refinement
- detecting physical inhomogeneities using impedance tomography, and
- synergizing tomography and performance modeling for concrete life prognosis.



Development of Non-Proprietary UHPC Concrete Mixtures Driven by LCA modeling

- Mobasher, Neithalath, Geers (Bekaert), Bonakdar (Euclid Chemicals), Schaefer (Masterbuilders Solutions)
- Life cycle cost and economic advantages of using UHPC systems in terms of structural applications.
- Precast UHPC elements reduce the overall operational complications, labor-intensive tasks. From a design perspective, elastoplastic behavior supports durability initiatives, ductility, allows load redistribution, and weight reduction.



Multi-scale material design and geometry optimization for 3D-printed multifunctional building envelopes

- **Team:** Narayanan Neithalath (ASU), Clarence McAllister (3D Construction)
- develop multifunctional 3D printable concrete mixtures for structural and thermal performance,
- Relationship between printability to large scale printing,
- design of lattice geometries given the constraints of materials, printability, and component cost
- Influence of material and geometry on heat transfer by concurrent optimization strategies for material and geometric design,
- large scale structural and thermal testing of building envelopes

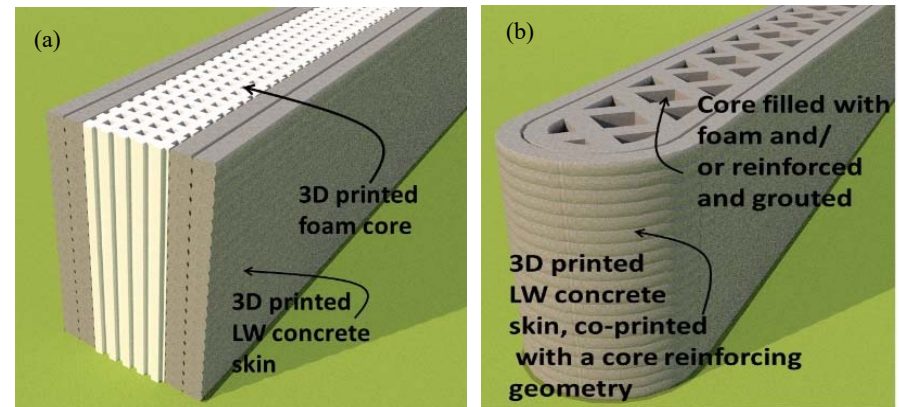
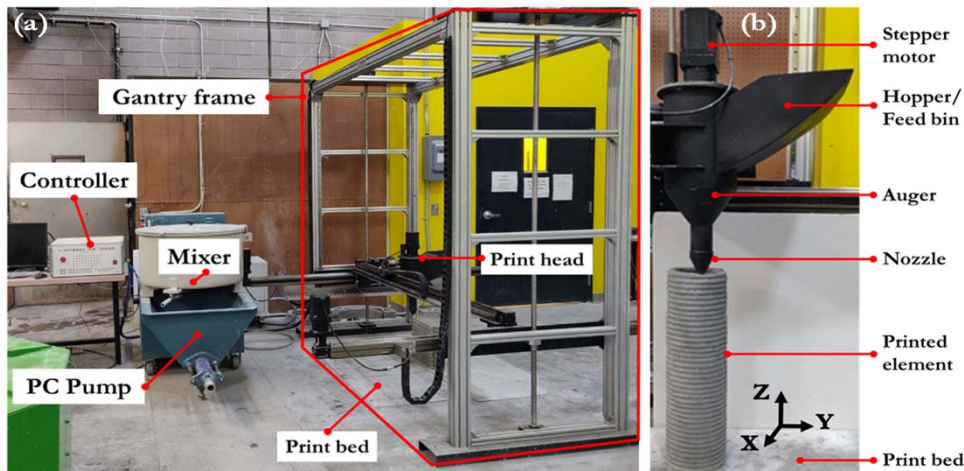


Fig. 7: Configurations of proposed, multi-material, 3D-printed, structurally and thermally efficient envelopes

Value Propositions

- Carbon footprint reduction: through use of clinker factor reduction (e.g., PLCs), SCMs, sequestration, upcycling, mineral carbonation
- Improving strength and durability of concrete: Construction practices, novel materials including UHPC, internal curing, crack control, extending service life, reducing repair cost
- Sourcing and evaluating alternate materials: Address fly ash shortage through performance establishment of reclaimed fly ash, evaluate natural pozzolan sources and properties
- Novel processing methods for cement and other materials: Alternate cements with lower energy and carbon footprint through fundamental changes in manufacturing, elimination of calcining in LC3 cement
- Predictive analytics in concrete: Prediction tools for service life, predictive methods for properties from big-data/machine learning (e.g., concrete performance prediction using RMC plant data, to optimize materials/processes)
- Economics of construction materials: enhance economy/reduce cost through optimization, waste minimization, enhanced recycling/reuse, better predictive tools

Value Propositions

- Enhancing construction speed: Developing and implementing fast setting/hardening materials, low-shrinkage/crack control, construction processes to accelerate construction including precast concrete and ABC
- Developing test methods for enhanced adoption of novel materials or robust detection of damage: performance-based tests,
- Hybrid Structural design tools using first principals to incorporate hybrid Supplementary Reinforcing Materials, SRM, such as nano fibers, micofibers, textiles, fibers, FRP, in conjunction with rebars to accomplish the serviceability, durability, and ductility.
- Digital construction: Faster, high-quality construction, materials design for 3D printing, strategies to incorporate reinforcement, reduce shrinkage, test methods and standards development to accelerate adoption
- Sensing and detection
- Self healing/self-sensing materials
- Design for serviceability and durability: enhanced adoption of materials such as textile reinforced concrete
- Tackling specific durability issues like AST/F-T/Sulfate attack: Devise appropriate, problem-specific actions, from the viewpoints of material selection, test methods, and service-life considerations
- Information/modeling: Science-based models for fundamental material properties/performance, so as to accelerate materials discovery

	VP	Approach
1	Carbon footprint reduction:	Clinker factor reduction (e.g., PLCs), SCMs, sequestration, upcycling, mineral carbonation
2	Improving strength and durability	Construction practices, novel materials including UHPC, FRC, internal curing, crack control, extending service life, reducing repair cost
3	Sourcing and evaluating alternate materials	Address fly ash shortage through performance establishment of reclaimed fly ash, evaluate natural pozzolan sources and properties, ASTM guidelines
4	Novel processing methods	Alternate cements with lower energy and carbon footprint through manufacturing process control, elimination of calcining in LC3 cement
5	Predictive analytics in concrete	Prediction tools for service life, properties from big-data/machine learning (e.g., concrete performance prediction using RMC plant data, to optimize materials/processes), QC
6	Economics of construction materials	enhance economy/reduce cost through optimization, waste minimization, enhanced recycling/reuse, better predictive tools
7	Enhancing construction speed:	Developing and implementing fast setting/hardening materials, low-shrinkage/crack control, Accelerate construction including precast concrete, ABC, Valley Metro, SRP, QC
8	Performance-based tests	Test methods for enhanced adoption of materials or robust detection of damage integrated in design
9	Hybrid Structural design tools	hybrid Supplementary Reinforcing Materials (SRM), nano /microfibers, textiles, fibers, FRP, in conjunction with rebars for serviceability, durability, and ductility.
10	Digital construction:	Faster, high-quality construction, materials design for 3D printing, incorporation of reinforcement
11	Sensing and detection	test methods and standards development to accelerate adoption
12	Self healing/sensing materials	Tools available to maintain serviceability through microstructure manipulation
13	Design for serviceability and durability:	Adoption of materials to address tangible design concerns such as crack width, stiffness, cycles of loading. Use of alternative materials for repair such as Textile reinforced Concrete
14	Tackling specific durability issues	ASR/F-T/Sulfate attack: Devise appropriate, problem-specific actions, from the viewpoints of material selection, test methods, and service-life considerations
15	Information/modeling	Science-based models for fundamental material properties/performance